

Andrew Joros

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SUMMARY

Research computing engineer specializing in AI/ML, cloud computing, and data engineering for environmental science applications. Over 12 years developing end-to-end analytical pipelines—from CFD simulations and machine learning models to AI-enhanced analysis and real-time monitoring dashboards—supporting climate research, water resource management, genomics, and wildfire modeling across federal, state, and academic partners.

EXPERIENCE

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| Research Computing Engineer (Software Development / Data Scientist) | 2024 – Current |
| <i>Division of Atmospheric Science / Division of Hydrologic Sciences, Desert Research Institute</i> | <i>Reno, NV</i> |
| <ul style="list-style-type: none">• Ported the USGS basin water model from Fortran to Python for rain-on-snow storm tests.• Cross-compared seven Nevada county water-plan drafts with AI and automated 262-basin pumping figures.• Track western snow drought each day on a public dashboard fed by mountain snow stations.• Classified western U.S. spring heatwaves with self-organizing machine learning on two historical weather datasets.• Converted five alpine wind-flow snow simulations into a fast machine-learning stand-in.• Combined storm shifts, hourly snow stations, and 7-day soil-moisture plots into a rain-on-snow dashboard.• Built a Nevada web tool for extreme design storms, watersheds, and area rainfall. | |
| Research Computing Engineer (Cloud Computing / Data Scientist) | 2019 – 2023 |
| <i>Center for Genomic Medicine & Climate Ecosystem Fire Applications, Desert Research Institute</i> | <i>Reno, NV</i> |
| <ul style="list-style-type: none">• Provided cloud computing and data engineering support for genetic analysis projects• Worked with team members on cloud-based data engineering and statistical projects, improving productivity• Led AWS onboarding and training, enhancing the team's cloud computing skills• Created AWS-based solutions for statistical smoke dispersion models for California's fire management• Developed a dynamic front-end webpage using AWS, presenting statistical modeling outputs related to Interagency Wildland Fire Air Quality, empowering stakeholders with valuable insights into air quality conditions | |
| Staff Research Scientist (Software Development / Data Scientist) | 2016 – 2019 |
| <i>Applied Innovation Center, Desert Research Institute</i> | <i>Reno, NV</i> |
| <ul style="list-style-type: none">• Developed water surveying software for groundwater assessment using vertical electrical sounding, enhancing well siting and drilling• Contributed to the construction of meteorological and air pollutant data pipeline, aiding environmental research for the Healthy Nevada Project• Integrated Christopher Ranch's sensor data into AWS for storage and machine learning use• Leveraged AWS EC2 to devise and implement cutting-edge agricultural insect and disease probabilistic models, empowering customers with accurate predictions for effective pest management and crop protection• Created an automated AWS sensor alert system for data integrity and reliability | |
| Assistant Research Scientist (High Performance Computing) | 2015 – 2016 |
| <i>Applied Innovation Center, Desert Research Institute</i> | <i>Reno, NV</i> |
| <ul style="list-style-type: none">• Constructed backend data architecture for the WINDS platform, improving weather data processing and decision support• Developed a web portal for NDOT, enabling engineers to access and analyze traffic data for transportation planning• Led NevCAN's data visualization project, presenting real-time climate and ecological data effectively• Created an image processing algorithm for greenhouse experiments to analyze vegetation health and growth | |
| Staff Scientist Programmer, Hydrometeorologist | 2013 – 2015 |
| <i>Desert Research Institute</i> | <i>Reno, NV</i> |
| <ul style="list-style-type: none">• Used Earth Engine JavaScript API for geospatial data analysis in hydrometeorology• Contributed to the Climate and Integrated Earth Monitoring Engine dashboard, enhancing environmental data visualization | |

- Processed Landsat and weather data via cloud computing, aiding Nevada's ecosystem analysis and conservation
- Evaluated precipitation datasets against Texas weather stations to improve irrigation and water management

Research Assistant

2009 – 2013

Desert Research Institute

Reno, NV

- Investigated monsoonal surges in the Northern Great Basin using statistical and dynamic methods
- Improved weather datasets to better represent irrigated agricultural environments
- Worked with CEFA to create a RAWS network density plot, aiding land managers' decisions
- Developed web frameworks for NOAA-funded drought monitoring and an El Nino climate risk project

Statistical Analyst

Summer 2009

Atmospheric Systems Corporation

Valencia, CA

- Conducted statistical quality assurance and quality control (QA/QC) of output data from Sonic Detection and Ranging (SODAR) systems

EDUCATION

University of Nevada

Reno, NV

M.S. in Atmospheric Science

2009 – 2011

San Jose State University

San Jose, CA

B.S. in Meteorology

2006 – 2009

SKILLS

Programming Languages: Python, MATLAB, R, Java, C/C++, FORTRAN, Scala
Scientific / ML: xarray, scikit-learn, MiniSom, OpenFOAM, ParaView, PyVista, ArcGIS
Software Frameworks: Apache Hadoop, Apache Spark, Cloudera CDH, Flask, Django
Web Design: HTML5, CSS, JavaScript, Highcharts, PHP, MySQL
Cloud Platforms: Amazon Web Services, Google Cloud Platform, Azure
Operating Systems: Linux (Ubuntu/CentOS/Redhat), Windows

PUBLICATIONS AND REPORTS

2026

1. Rhoades, A. M., North, J. S., Rudisill, W., Hatchett, B. J., Risser, M., Beltran-Peña, A., Heggli, A., Hotaling, S., Huning, L. S., Joros, A., LaPlante, M., Mahesh, A., Marshall, A. M., McCrary, R., McEvoy, D. J., Rahimi, S., Raleigh, M. S., Randall, C., Srivastava, A., Wehner, M., Zhou, Y., & Jones, A. D. (2026). Snow-eater heat waves of the western United States. *Science Advances*, 12(32), eab3361. AAAS, August 5, 2026. <https://doi.org/10.1126/sciadv.aeb3361>
2. Yu, G., Miller, J. J., Mejia, J. F., Rasouli, K., McCarthy, M. I., Joros, A. N., & O'Shea, K. (2026). Updating the Probable Maximum Precipitation for Nevada. Division of Hydrologic Sciences, Desert Research Institute, DHS Publication #41312, prepared for the Nevada Division of Water Resources, June 30, 2025.

2020

3. Kiser, D., Metcalf, W. J., Elhanan, G., Schnieder, B., Schlauch, K., Joros, A., Petersen, C., & Grzyski, J. (2020). Particulate matter and emergency visits for asthma: a time-series study of their association in the presence and absence of wildfire smoke in Reno, Nevada, 2013–2018. *Environmental Health*, 19, 92. BioMed Central, August 27, 2020. <https://doi.org/10.1186/s12940-020-00646-2>

2017

4. Caldwell, T. G., Huntington, J. L., Scanlon, B., Joros, A. N., & Howard, T. (2017). Improving Irrigation Water Use Estimates with Remote Sensing Technologies: An Initial Feasibility Study for Texas, 137 p. University of Texas, Bureau of Economic Geology: Austin, TX. Report prepared for the Texas Water Development Board.

2015

5. Huntington, J. L., Gangopadhyay, S., Spears, M., Allen, R., King, D. L., Morton, C. G., Harrison, A., McEvoy, D. J., & Joros, A. N. (2015). West-Wide Climate Risk Assessments: Irrigation Demand and Reservoir Evaporation Projections. Technical Memorandum No. 68-68210-2014-01, 218 p., 754 appx. U.S. Bureau of Reclamation.

2013

6. Brown, T., Larkin, N. K., Rorig, M., & Joros, A. (2013). Improving Meteorological Data and Forecasts for Prescribed Fire Burn Day Decisions for the Lake Tahoe Basin. CEFA Final Report (SNPLMA).

2011

7. Joros, A. (2011). Extratropical Control of Monsoonal Surges in the Great Basin. Master's thesis, University of Nevada, Reno, NV.

2009

8. Joros, A., & Abatzoglou, J. (2009). Analysis of Monsoonal Surges into Southern California Deserts. Senior thesis, San Jose State University, San Jose, CA.

CONFERENCE PAPERS/PRESENTATIONS

1. Saftner, D. M., Arienzo, M. M., Atkin, J., Daniel, A. T., Dunkerly, C. W., Gordon, B., Greene, S. J., Huntington, J. L., Joros, A. N., Kozloski, R., Lutz, A. D., Parker, B. A., ReVelle, P., & VanderMolen, K. A. (2026). Developing Water Resource Plans for Rural Nevada. Nevada Tribal Food and Agriculture Summit: Sparks, NV, March 24–27, 2026, Invited.
2. Yu, G., Miller, J. J., Mejia, J. F., Rasouli, K., McCarthy, M. I., O'Shea, K., & Joros, A. N. (2025). Updating the Probable Maximum Precipitation for Nevada. 2025 NWRA Annual Conference: Sparks, NV, January 27–30, 2025.
3. McEvoy, D. J., Rhoades, A., Huning, L., Heggli, A. E., Hatchett, B., & Joros, A. N. (2025). Storm-scale impacts on seasonal snow droughts: A case study in the Pacific Northwest, United States. American Geophysical Union Annual Meeting: New Orleans, LA, December 17, 2025.
4. McEvoy, D. J., Rhoades, A., Huning, L., Heggli, A. E., Hatchett, B., & Joros, A. N. (2025). Storm-scale impacts on seasonal snow droughts: A case study in the Pacific Northwest, United States. International Mountain Conference: Innsbruck, Austria, September 14–18, 2025.
5. Hatchett, B., Abatzoglou, J., & Joros, A. (2019). Monsoon-driven variability in fire-climate linkages. Western Governors' Association Fire & Climate Science meeting.
6. Rowan, C., Metcalf, W. J., Elhanan, G., Joros, A. N., & Grzymiski, J. J. (2018). Short-term sentinel air events in relation to health care utilization for specific health conditions in Reno, Nevada. American Thoracic Society International Conference, B24, A2815.
7. Rosenquist, N. A., Metcalf, J., Elhanan, G., Grzymiski, J., Joros, A. N., Field-Ridley, A., & Darrow, L. A. (2018). Acute associations between PM_{2.5} and allergic and nonallergic asthma exacerbations in children and adults. ISEE Conference Abstracts, 2018(1).
8. Huntington, J. L., Daudert, B., Morton, C. G., McEvoy, D. J., Joros, A. N., Abatzoglou, J. T., Hegewisch, K., Peterson, A., Vansant, D. S., Allen, R., Kilic, A., Hobbins, M., & Verdin, J. (2015). Cloud computing for drought monitoring with Google Earth Engine. U.S. Drought Monitor Forum: Desert Research Institute, Reno, NV, April 14–16, 2015, Invited.
9. Hegewisch, K., Daudert, B., Morton, C. G., Peterson, A., Joros, A. N., McEvoy, D. J., Erickson, T., Huntington, J. L., & Abatzoglou, J. T. (2015). Google drought monitoring through cloud computing and visualization of remote sensing and meteorological datasets: Examples for California. American Geophysical Union Chapman Conference: Irvine, CA, April 20–22, 2015.
10. Huntington, J. L., Morton, C. G., McGwire, K. C., Allen, R., Gorelick, N., Thau, D., & Joros, A. N. (2014). Monitoring Groundwater Dependent Ecosystems in the Great Basin from Space and Clouds. American Water Resources Association Summer Specialty Conference: Reno, NV, May 31–July 2, 2014.
11. Huntington, J. L., Morton, C. G., McGwire, K. C., Joros, A. N., Peterson, S., Gorelick, N., Thau, D., & Allen, R. (2014). Cloud Computing of Landsat Imagery and Gridded Weather Data for Evaluating Groundwater Dependent Ecosystems in Nevada. Nevada Water Resources Association Annual Conference: Las Vegas, NV, February 2, 2014.
12. Huntington, J. L., Morton, C. G., McGwire, K. C., Joros, A. N., Peterson, S., Gorelick, N., Thau, D., & Allen, R. (2014). Utilizing Cloud Computing of Landsat Imagery and Gridded Weather Data for Evaluating Groundwater Dependent Ecosystems in Nevada. Nevada Water Resources Association Annual Conference: Las Vegas, NV, February 3–6, 2014.
13. Allen, R. G., Huntington, J. L., Kilic, A., Debruin, H., & Joros, A. (2013). Conditioning of NLDAS, NARR and arid weather station data to improve their representation of well-watered (reference) environments associated with irrigated agriculture. 93rd AMS Annual Meeting, 27th Conference on Hydrology: Austin, TX, January 5–10, 2013.
14. Redmond, K. T., Abatzoglou, J. T., McEvoy, D., Joros, A., VanSant, D., & Edwards, L. M. (2013). The WestWide Drought Tracker: Drought Monitoring at Fine Spatial Scales. 93rd AMS Annual Meeting: Austin, TX, January 5–10, 2013.
15. Nauslar, N., Abatzoglou, J., & Joros, A. (2011). Development of an objective methodology for diagnosing Santa Ana winds. Ninth Symposium on Fire and Forest Meteorology: Palm Springs, CA, October 18–20, 2011.

16. Joros, A., & Mejia, J. F. (2011). Impact of Eastern and Central Pacific ENSO events over the Continental United States. AMS Applied Climatology Conference: Asheville, NC, July 18–20, 2011.
17. Uher, E., Kaplan, M. L., Joros, A., & Decker, D. (2011). Air pollution dispersion forecasting: A climatological study of Cape Canaveral tropospheric wind patterns. 91st AMS Annual Meeting: Seattle, WA, January 23, 2011.
18. King, K. C., Kaplan, M. L., Joros, A., Liddle, M., & Uher, E. (2011). Evaluation of the Operational Multi-scale Environment model with Grid Adaptivity (OMEGA) for use in Wind Energy Potential Assessment in the Great Basin of Nevada. 91st AMS Annual Meeting: Seattle, WA, January 23–27, 2011.
19. Abatzoglou, J. T., Redmond, K. T., Edwards, L. M., & Joros, A. (2011). Monitoring of drought in the western United States: metrics suitable for tracking drought historically and throughout the 21st century. 91st AMS Annual Meeting: Seattle, WA, January 23–27, 2011.
20. Joros, A., Abatzoglou, J., Favors, J., Tan, B., & Kaplan, M. (2010). Extratropical Control of Monsoonal Surges into the Great Basin. AMS Mountain Meteorology Conference: Lake Tahoe, CA, August 30–September 3, 2010.
21. Tan, B., Joros, A., & Kaplan, M. L. (2010). The Role of Midlatitude Circulations in Triggering Extratropical Convection during the 2004 North American Monsoon. 90th AMS Annual Meeting: Atlanta, GA, January 17–22, 2010.
22. King, K. C., Kaplan, M., Joros, A., Liddle, M., & Uher, E. (2010). Evaluation of the Operational Multiscale Model with Grid Adaptivity (OMEGA) for use in Wind Energy Potential Assessment in the Great Basin of Nevada. AMS Mountain Meteorology Conference: Lake Tahoe, CA, August 30–September 3, 2010.